

REQUIREMENT ANALYSIS

Metro Ticket Generating System in ServiceNow

Requirement Analysis

This phase converts the ideation outcomes into clear customer needs, functional and non-functional requirements, user stories, data movement, and the technical stack required for the ServiceNow solution.

1. Customer Journey Map

The customer journey is mapped for a metro passenger using the proposed digital booking service.

Journey Stage	Customer Action	Touchpoint	Pain Point	Expected Experience	System Opportunity
Need to travel	Decides source and destination	Metro service	Needs quick ticket access	Clear and simple booking	Service Catalog entry
Start booking	Opens Book A Metro Ticket	Service Portal	Manual queues may take time	Simple digital form	Catalog item
Enter journey details	Selects starting/going-to stations, journey type and passengers	Booking form	Fare can be confusing	Fare is calculated automatically	Client-side fare logic
Payment	Chooses UPI/Card/Others	Payment section	Payment method may vary	Relevant payment field appears	Catalog UI Policy
Confirm	Reviews order	Order confirmation	Wants confidence before submitting	Clear confirmation	Catalog workflow
Receive ticket	Gets booking/request and QR	ServiceNow / QR modal	Needs an accessible ticket	Digital QR ticket	QR widget
Record/track	Ticket is stored for operational use	Metro Database	Manual record maintenance	Centralized record	Flow Designer

2. Solution Requirements (Functional & Non-functional)

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Metro Ticket Booking	Provide a Service Catalog item named Book A Metro Ticket.
FR-2	Station Selection	Allow Starting From and Going

		To station selection using Metro Station's Details reference data.
FR-3	Journey Selection	Support Single journey and Return journey.
FR-4	Passenger Selection	Allow 1–4 passengers.
FR-5	Fare Calculation	Automatically calculate fare from source, destination, journey type and passenger count.
FR-6	Payment	Support UPI, Card and Others.
FR-7	Conditional Payment Details	Show Enter Payment Mode when Others is selected.
FR-8	QR Ticket	Generate a QR representation for the ticket.
FR-9	Ticket Record	Create a Metro Database record containing journey, fare, payment and QR information.
FR-10	RITM Tracking	Store a readable RITM Number with the ticket record.
FR-11	Validation	Prevent the same station from being selected as source and destination.
FR-12	Automation	Use Flow Designer to map catalog variables into the Metro Database.

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The booking form should be simple and understandable for passengers.
NFR-2	Security	Use ServiceNow roles/ACLs and platform access controls for custom data.
NFR-3	Reliability	Fare calculations and record creation should be consistent for valid inputs.
NFR-4	Performance	Form actions and automated record creation should complete without unnecessary delay.
NFR-5	Availability	The solution should use the availability characteristics of the ServiceNow platform.
NFR-6	Scalability	Master station data and ticket records should support future expansion.

3. Data Flow Diagram & User Stories

DFD — Level 1 textual representation

Entity / Process	Data Flow	Data Store / Output
Passenger	Station, journey type, passenger count, payment choice	Book A Metro Ticket catalog form
Catalog Form	Booking details → fare logic	Fare fields
Fare Logic	Source + destination + journey + passengers	Single/return amount
Payment UI Policy	Mode of Payment = Others	Enter Payment Mode visibility
QR Generation	Ticket identifier / URL → QR image	QR widget / digital ticket
Flow Designer	Catalog variables + Requested Item	Metro Database
Metro Database	Ticket, RITM, route, fare, payment, QR status	Operational ticket record

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Passenger	Ticket Booking	USN-1	As a passenger, I can open the metro ticket catalog item and enter my journey details.	Booking form is accessible and accepts valid details.	High	Sprint-1
Passenger	Station Selection	USN-2	As a passenger, I can select starting and destination stations.	Both station fields provide valid station records.	High	Sprint-1
Passenger	Journey Type	USN-3	As a passenger, I can choose single or return journey.	Selected journey type is stored and used in fare calculation.	High	Sprint-1
Passenger	Passenger Count	USN-4	As a passenger, I can select the number of passengers.	Values 1–4 are available.	High	Sprint-1

Passenger	Fare Calculation	USN-5	As a passenger, I can see the calculated fare based on my selections.	Correct single/return amount is populated.	High	Sprint-1
Passenger	Payment	USN-6	As a passenger, I can choose UPI, Card or Others.	Payment option is stored; Others reveals its additional field.	High	Sprint-2
Passenger	QR Ticket	USN-7	As a passenger, I can receive a QR representation of my ticket.	QR is displayed in the ticket flow.	High	Sprint-2
Operations	Ticket Records	USN-8	As an operations user, I can view ticket records in Metro Database.	Ticket records contain required booking information.	High	Sprint-2
Operations	RITM Tracking	USN-9	As an operations user, I can identify the related RITM number.	Readable RITM Number is stored.	Medium	Sprint-2
Admin	Access Control	USN-10	As an administrator, I can control access to metro station data.	Applicable table ACLs/roles protect the custom table.	High	Sprint-2

4. Technology Stack (Architecture & Stack)

Technical architecture is centered on ServiceNow, using Service Catalog for the passenger-facing request, client scripting for fare/validation logic, Flow Designer for automation, custom tables for master and transactional data, and a QR service/widget for digital ticket presentation.

S.No.	Component	Description	Technology
1	User Interface	Passenger booking form and confirmation	ServiceNow Service Portal / Service Catalog
2	Application Logic-1	Fare calculation and same-station validation	Catalog Client Script / JavaScript

3	Application Logic-2	Conditional display of payment details	Catalog UI Policy
4	Application Logic-3	Automated record creation and field mapping	Flow Designer
5	Database	Station master data	ServiceNow custom table: u_metro_station_details
6	Database	Ticket/transaction records	ServiceNow custom table: u_metro_database
7	External API	QR image generation	QRServer API used by the QR URL
8	Digital Ticket UI	QR presentation	ServiceNow Service Portal widget
9	Security	Access control	ServiceNow roles and ACLs
10	Infrastructure	Application platform	ServiceNow cloud platform

Application Characteristics

S.No.	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the ServiceNow implementation.	ServiceNow platform components
2	Security Implementations	Custom station table access is protected by ServiceNow ACLs/roles; platform authentication and authorization apply.	ServiceNow ACL / RBAC
3	Scalable Architecture	Station master data, catalog variables, flows and transaction records are separated by responsibility.	ServiceNow tables + Flow Designer
4	Availability	Hosted ServiceNow platform provides the application runtime.	ServiceNow
5	Performance	Client-side fare calculation avoids unnecessary server calls for every selection; automated flow handles record creation.	JavaScript + Flow Designer

Requirement Analysis Outcome

The requirements establish a traceable path from passenger need → catalog booking → fare calculation → payment → QR ticket → automated Metro Database record.